❤️ BIOLOGY CH.9 — CIRCULATORY SYSTEM — FULL MIND MAP

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📄 Topics in this chapter:

1. Human Heart Structure & Chambers

- Human heart 4-chambered hai: 2 upper Atria (Auricles) + 2 lower Ventricles. Left ventricle ki walls sabse moti hoti hain kyunki ye poora body mein blood pump karta hai high pressure mein. *[Q537, Q553, Q598]*
- Septum heart ke left aur right sides ko completely alag rakhta hai taaki oxygenated aur deoxygenated blood mix na ho. *[Q537]*
- Heart valves (one-way inlets/outlets) backflow rokte hain: - Tricuspid valve: Right atrium ↔ Right ventricle - Pulmonary valve: Right ventricle ↔ Pulmonary artery - Aortic valve: Left ventricle ↔ Aorta - Mitral (Bicuspid) valve: Left atrium ↔ Left ventricle *[Q559]*
- Cardiac muscles cylindrical, branched, single-nuclei hoti hain, striated appearance hai, aur involuntary hain. *[Q593]*
- Purkinje fibres heart ki muscle fibres hain jo electrical stimulus conduct karti hain taaki coordinated heart contraction ho sake. Jan Evangelista Purkyne ne 1839 mein discover kiya. *[Q577]*
- Heart weight adult mein ~300 gm hota hai (range 236-412 gm), size ek closed fist ke barabar. *[Q571]*

2. Heartbeat Cycle - Systole & Diastole

- Diastole (relaxation): Blood ventricles mein enter karta hai. Ventricular diastole mein oxygenated blood left atrium → left ventricle; deoxygenated blood right atrium → right ventricle. *[Q532]*
- Systole (contraction): Blood ventricles se bahar nikalta hai. Systolic BP ventricular contraction ke dauran measure hota hai. *[Q535]*
- Pulse arteries mein blood flow ki rhythmic throbbing hai - arteries ki muscular walls expand hoti hain jab heart se blood pump hota hai. *[Q558]*
- Normal BP: <120/80 mmHg. Systolic = max pressure (heart beats), Diastolic = min pressure (heart rests). High BP → Hypertension + kidney problems. Low BP → Dizziness + Nausea. *[Q535, Q544, Q572]*

3. Blood Vessels - Arteries, Veins, Capillaries

- Arteries: Thick walls, no valves, blood high pressure mein bahata hai, heart se body ke parts mein blood le jaati hai (except pulmonary artery jo deoxygenated blood lungs tak leti hai). *[Q545, Q582]*
- Veins: Thin walls, valves hoti hain, low pressure, body se blood heart ki taraf leti hain (except pulmonary vein jo oxygenated blood lungs se heart tak leti hai). Skin ke paas dikhti hain. *[Q545, Q552]*
- Capillaries: Sabse chhoti vessels, arteries ko veins se connect karti hain, tissues tak O_2/nutrients supply karti hain. Skin ke neeche maujood chhoti vessels capillaries hain. *[Q573, Q581]*
- Aorta: Sabse badi artery, left ventricle se oxygenated blood body ke baaki hisson mein leti hai. *[Q562, Q566]*
- Vena Cava: Sabse badi vein, deoxygenated blood heart tak leti hai. *[Q566]*
- Pulmonary vein unique hai kyunki ye ekmatr vein hai jo oxygenated blood carry karti hai (lungs → left atrium). *[Q543, Q561, Q579]*
- Renal artery heart se kidney tak blood leti hai. Coronary arteries heart muscle ko khud oxygenated blood supply karti hain. *[Q570]*

4. Blood Pressure & Measurement

- BP measure karne ke liye Sphygmomanometer use hota hai. *[Q540]*
- Stethoscope heartbeat detect karne mein help karta hai. Thermometer temperature, Spirometer breath amount measure karta hai. *[Q542]*
- ⚠️ Trap: Hypertension (High BP) veins ke constriction se nahi hota - ye arteries/arterioles ke constriction se hota hai. *[Q534, Q551]*
- BP = force jo blood vessel ki wall ke against exert hota hai. Normal range 120/80 se 140/90 mmHg tak. *[Q572]*

5. Double vs Single Circulation

- Double circulation: Ek cycle mein blood heart se do baar guzarta hai. - Complete double: Birds (Peacock, Pigeon), Mammals (Man, Tiger, Rabbit, Bat, Whale), Crocodiles - 4-chambered heart. - Incomplete double: Amphibians (Frog, Salamander), most Reptiles - 3-chambered heart, thoda mixing tolerate kar sakte hain. *[Q533, Q538, Q541, Q546, Q554, Q557, Q563, Q584]*
- Single circulation: Fish mein - ek cycle mein blood heart se ek baar guzarta hai. 2-chambered heart (1 atrium + 1 ventricle). *[Q538, Q547, Q548, Q554, Q568, Q583]*
- Deoxygenated blood ko alag rakha jata hai taaki warm-blooded animals (birds, mammals) ka body temperature constant rahe - cellular respiration ke liye zyada oxygen chahiye hoti hai. *[Q549]*

6. Heart Chambers in Different Animals

- 2-chambered: Fish, Seahorse (1 atrium + 1 ventricle) *[Q541, Q548, Q583]*
- 3-chambered: Amphibians (Frog, Salamander), most Reptiles (2 atria + 1 ventricle) - mixing tolerate kar sakte hain. *[Q541, Q546, Q584]*
- 4-chambered: Birds (Peacock, Pigeon), Mammals (Man, Tiger, Rabbit, Bat, Whale), Crocodiles - complete separation. *[Q541, Q546, Q548, Q563]*
- Cockroach: 13-chambered heart. *[Q583]*

7. Blood Components & Functions

- RBC (Erythrocytes): Oxygen carry karte hain lungs se body tak. Biconcave disc shape, diameter ~7.5 μm, lifespan 115-120 days, nucleus nahi hota. Hemoglobin contain karte hain. Bone marrow mein bante hain. *[Q574, Q592, Q594, Q595]*
- WBC (Leukocytes): Immunity, "soldiers" of the body. Blood ka sirf 1% hissa. Types: Granulocytes (Neutrophils, Eosinophils, Basophils), Monocytes, Lymphocytes (T cells, B cells). Normal count: 4,500-11,000 per microliter. *[Q536, Q560, Q596]*
- Platelets (Thrombocytes): Blood clotting (coagulation) mein help karte hain, bone marrow se bante hain. *[Q534, Q555]*
- Plasma: Blood ka liquid part, 55% of total blood volume. 92% water + 8% proteins/minerals/hormones/enzymes. Food, CO_2, nitrogenous waste transport karta hai. *[Q599, Q600]*
- Blood ka function: O_2 carry, CO_2 remove, immunity (WBC), clotting (platelets). ⚠ Trap: Blood sensory inputs mein help nahi karta - ye nervous system ka kaam hai. *[Q569]*

8. Hemoglobin & Iron

- Hemoglobin iron-rich protein hai RBCs mein jo O_2 bind karke transport karta hai. *[Q567]*
- Hemoglobin do parts se bana: Heme (iron part, red colour) + Globin (protein part). *[Q564, Q576]*
- Iron deficiency se Anaemia hota hai. *[Q564]*

9. Blood Groups - ABO & Rh

- 8 main blood types: A+, A-, B+, B-, O+, O-, AB+, AB-. *[Q539]*
- Universal Donor: O (O- ideally, O+ emergency mein positive types ke liye) - RBC par na A na B antigen hota, dono antibodies plasma mein hote hain. *[Q539]*
- Universal Recipient: AB+ - dono A aur B antigens hote hain, koi antibodies nahi hoti. *[Q565]*
- A + B parents se AB child = Co-dominance example. A aur B dono genes dominant hain. *[Q556]*

10. Blood Clotting (Coagulation)

- Platelets clotting ke liye crucial hain - injury site par clot form karte hain excess blood loss rokne ke liye. *[Q555]*
- Calcium (Ca, atomic no. 20) blood coagulation ke liye essential hai - platelets aur plasma proteins ke saath interact karke clot banata hai. *[Q578]*

11. Lymphatic System

- Lymph plasma jaisa hai lekin colorless aur kam protein hota hai. Isme WBCs hote hain jo harmful bacteria se ladte hain. Tissue fluid bhi kehte hain. Digested food aur fat transport karta hai. *[Q534, Q550]*

12. Blood as Connective Tissue

- Blood ek red vascular connective tissue hai - fluid matrix (plasma) + formed elements (RBC, WBC, platelets). *[Q585, Q591]*
- Blood body weight ka 7-8% hota hai. Adult mein ~5-6 litres (kuch questions mein 6.8L mentioned). pH 7.35-7.45 (slightly basic). *[Q569, Q586, Q587, Q588]*
- ⚠ Trap: Blood white nahi hota - ye exclusively red fluid connective tissue hai. *[Q586, Q587]*
- Blood mein fibers absent hote hain - ye specialized liquid connective tissue hai. *[Q597]*
- ⚠ Trap: Blood flow jerkily aur low pressure par nahi hota - ye smooth aur steady hota hai, arteries mein high pressure. *[Q589]*
- ⚠ Trap: Vascular system body weight ka 1.7-1.8% nahi - ye 7-8% hota hai. *[Q588]*

13. Diagnostic Instruments

- Sphygmomanometer: Blood pressure measure karta hai. *[Q540]*
- Stethoscope: Heartbeat detect karta hai. *[Q542]*
- ECG (Electrocardiogram): Heart problems diagnose karta hai - irregularities in heart rhythm. Willem Einthoven ne invent kiya. Heart attack, heart failure, arrhythmia, cardiac arrest detect kar sakta hai. *[Q575]*
- Spirometer: Breath amount measure karta hai. *[Q542]*
- Haemocytometer: RBC count determine karta hai. *[Q595]*
- Spectrophotometer light measure karta hai, Urinometer urine specific gravity, Haemoglobin meter Hb level. *[Q540]*

14. Heart Facts & Other Important Points

- Angioplasty heart par ki jati hai - blocked coronary arteries ko open karne ke liye. *[Q580]*

- ECG, Stethoscope, Sphygmomanometer heart se related hain. EEG brain activity ke liye, BCG TB vaccine, ECT electroconvulsive therapy. *[Q575]*
- Human heart ~72 times/minute dhadata hai. Brain weight ~1300-1400 gm. *[Q580, Q571]*
- ⚠️ Disputed: Vascular system ke baare mein diye gaye statements mein — 'human heart closed fist ke size ka hai' wale option ko is specific question mein 'NOT true' maana gaya (halaki commonly heart ko fist-sized hi approximate kiya jaata hai) — baaki statements (blood body weight ka 7-8%, high pressure flow, 'river of life') sahi hain. *[Q590]*